

3. Have return history available for bad times

Factor risk premiums exist because they reward the willingness to suffer losses during bad times. Having some data points to measure worst-case scenarios is necessary for assessing risk–return trade-offs and risk management (see below). We also want a reasonable length of data to perform these exercises.

4. Be implementable in liquid, traded instruments

As in Canada Pension Plan Investment Board’s (CPPIB) Reference Portfolio, factor strategies should be dirt cheap, something best accomplished by constructing them from liquid securities. Scalability is an important requirement for large investors.

Factor strategies often involve leverage. You need to overweight value stocks and simultaneously underweight or short growth stocks. Going long value and short growth is a *dynamic leverage* strategy. Even if shorting is not possible, factor strategies still work: Israel and Moskowitz (2013) show that there are still significant value and momentum factor premiums available even if an investor cannot short, but the profitability of these factor strategies is dented by 50% to 60%.

Limiting ourselves to liquid securities might make it seem impossible to construct a factor illiquidity premium, but this is not so. You can collect an illiquidity premium by taking long positions in (relatively) illiquid assets and short positions in (relatively) liquid assets, all in liquid markets. It turns out that illiquidity premiums within (liquid) asset classes are much larger than illiquidity premiums across asset classes involving private equity, hedge funds, and other illiquid assets (see chapter 13). These types of illiquid assets also present tricky principal-agent problems and can be accessed only in murky markets where scant information is available.

The four criteria deliberately exclude the trendy factor du jour. A factor should be considered for a benchmark only if it is widely recognized. A factor that predicts returns by some statistical analysis of social networking data (yes, there are some, see chapter 8) is best left to active management. Many of the factors that we recognize today—like value-growth and momentum—were originally labeled as anomalies, or alpha, as they could not be explained by existing risk factors. As the literature matured and there was growing acceptance of these strategies by institutional investors, they became beta, or were treated as factor premiums in their own right.

Factors Can Appear and Disappear

The set of factors is not static. Options trading on centralized markets was introduced in the late 1960s, and these markets blossomed after the seminal option pricing formula of Black and Scholes (1973). Before then, it was not possible to obtain large-scale exposure to volatility risk. The high yield bond market rose to prominence in the late 1970s and 1980s, due largely to the financier Michael